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Two-Tower Recommender

Author	Document date	Validation
Sai Veda	May 30, 2023	33 automated tests

Abstract

This project documents a recommendation stack where retrieval and ranking are deliberately separated so candidate generation can scale without giving up downstream business relevance.

Project Background

Recommendation quality breaks when systems either over-index on popularity or overfit a ranking model that cannot scale to a full catalog. The architecture therefore mirrors production: first recall, then precision.

Headline Result

beats popularity +76% recall; ranker +7.8% nDCG

Scope Overview

Temporal data split and training statistics. Two-tower retrieval model and item embedding space. ANN candidate recall plus gradient-boosted reranking. Offline evaluation focused on recall and nDCG gains.

Project Context

A complete, offline, deterministic recommender system: a from-scratch two-tower retrieval model, ANN candidate generation, a gradient-boosted ranker, and a leak-free temporal evaluation harness - trained on millions of implicit interactions and architected for 1B. No GPU, no network, no paid APIs.

Executive Summary

Background, objectives, scope, and project impact.

Project Background

Recommendation quality breaks when systems either over-index on popularity or overfit a ranking model that cannot scale to a full catalog. The architecture therefore mirrors production: first recall, then precision.

Executive Summary

This project documents a recommendation stack where retrieval and ranking are deliberately separated so candidate generation can scale without giving up downstream business relevance.

Objectives

- Train user and item towers from scratch on implicit interactions.
- Use ANN retrieval to scale candidate generation.
- Improve final ranking quality with a second-stage ranker.

Scope

- Temporal data split and training statistics.
- Two-tower retrieval model and item embedding space.
- ANN candidate recall plus gradient-boosted reranking.
- Offline evaluation focused on recall and nDCG gains.

Impact

The combination of two-tower retrieval, ANN candidate generation, and a ranker gives the project a realistic online shape. The measured uplift over popularity and retrieval-only ranking makes the business value concrete.

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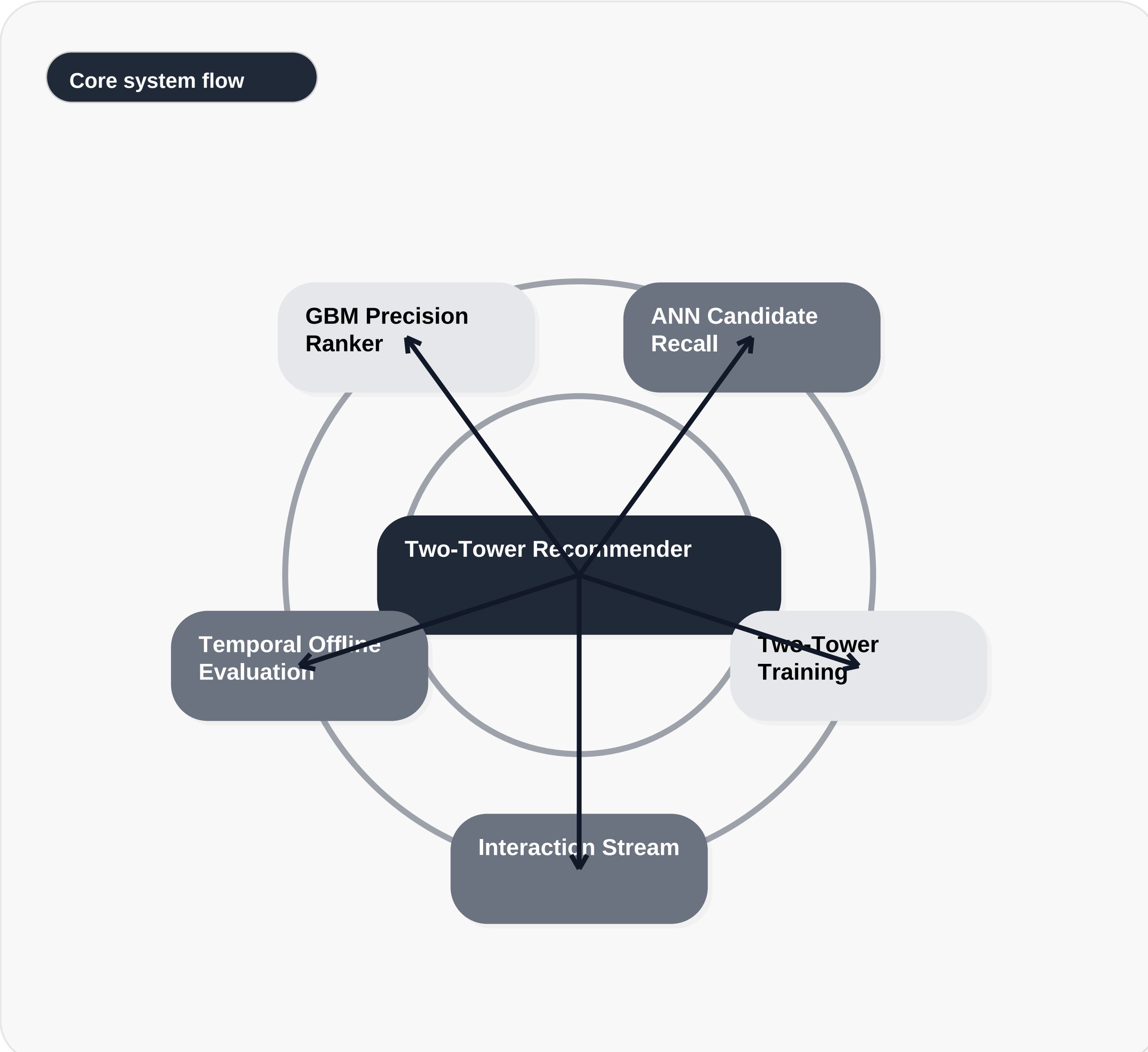
Solution Architecture

Candidate recall and ranking precision are intentionally split so the serving path scales without giving up business relevance.



Design Note 1

Retrieval and ranking are separated intentionally because they solve different scale problems.

Design Note 2

Temporal evaluation is non-negotiable for recommendation systems that serve future behavior.

Design Note 3

The item embedding space is treated as infrastructure that downstream ranking can build on.

Implementation Process

Delivery phases, engineering approach, and quality checkpoints.

Delivery Narrative

A complete, offline, deterministic recommender system: a from-scratch two-tower retrieval model, ANN candidate generation, a gradient-boosted ranker, and a leak-free temporal evaluation harness - trained on millions of implicit interactions and architected for 1B. No GPU, no network, no paid APIs.

1. **Interaction modeling** — Started with a temporal split so evaluation reflects forward-looking recommendation behavior.
2. **Embedding training** — Trained user and item towers with in-batch negatives and popularity correction.
3. **Two-stage serving path** — Separated ANN recall from a richer ranking stage that can use additional features.
4. **Outcome review** — Measured recall lift over popularity and ranking lift over retrieval-only baselines.

Quality Checkpoints

- 33 automated tests validate core behavior and edge cases.
- Benchmark files and screenshots are generated from committed project artifacts.
- The run path stays deterministic so numbers can be reproduced and reviewed directly.

Engineering Principles

- Retrieval and ranking are separated intentionally because they solve different scale problems.
- Temporal evaluation is non-negotiable for recommendation systems that serve future behavior.
- The item embedding space is treated as infrastructure that downstream ranking can build on.

Evidence And Results

Test evidence, benchmark interpretation, and screenshot review.

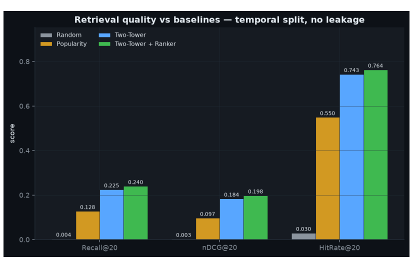
33 passing tests | **May 30, 2023** | **beats popularity +76% recall; ranker +7.8% nDCG**

Benchmark Evidence

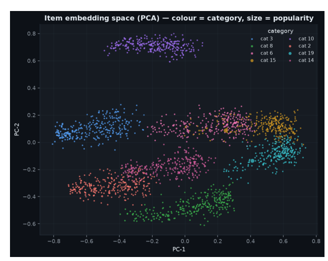
This repository emphasizes visual benchmark artifacts and automated validation outputs. Where tables are not present, the screenshots and test fixtures remain the primary review surface.

Result Interpretation

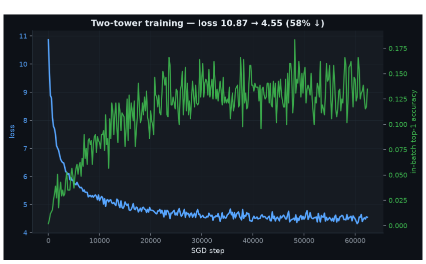
- The retrieval gains matter because they increase the candidate pool quality before ranking begins.
- The ranker lift is easier to explain when shown as an improvement over a strong retrieval baseline.
- The dashboard evidence makes clear that recall and nDCG improve for different reasons and should be



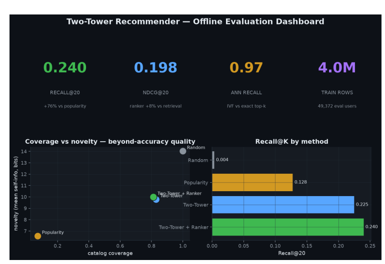
Retrieval quality vs baselines



Item embedding space (PCA), coloured by category



Two-tower training curve



Offline evaluation dashboard

Risks And Next Steps

Operational considerations, mitigation choices, and forward path.

Risk Register

Catalog scale	Addressed by using ANN for candidate recall instead of full-catalog scoring.
Evaluation leakage	Mitigated through a strict temporal split and feature discipline.
Over-reliance on popularity	Reduced by combining learned embeddings with richer ranking features.

Next Steps

- Add session-aware or context-aware retrieval features for short-term intent shifts.
- Experiment with harder negative sampling and refreshed item embeddings.
- Introduce business-constraint reranking for diversity, margin, or inventory goals.

Closing Note

The project brief is intentionally written as delivered work rather than a transfer note. It documents the business problem, the engineering choices, the measured evidence, and the remaining operational path in a format suitable for portfolio review.

Sai Veda

Project brief date: May 30, 2023

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